

Global CO₂ Emissions

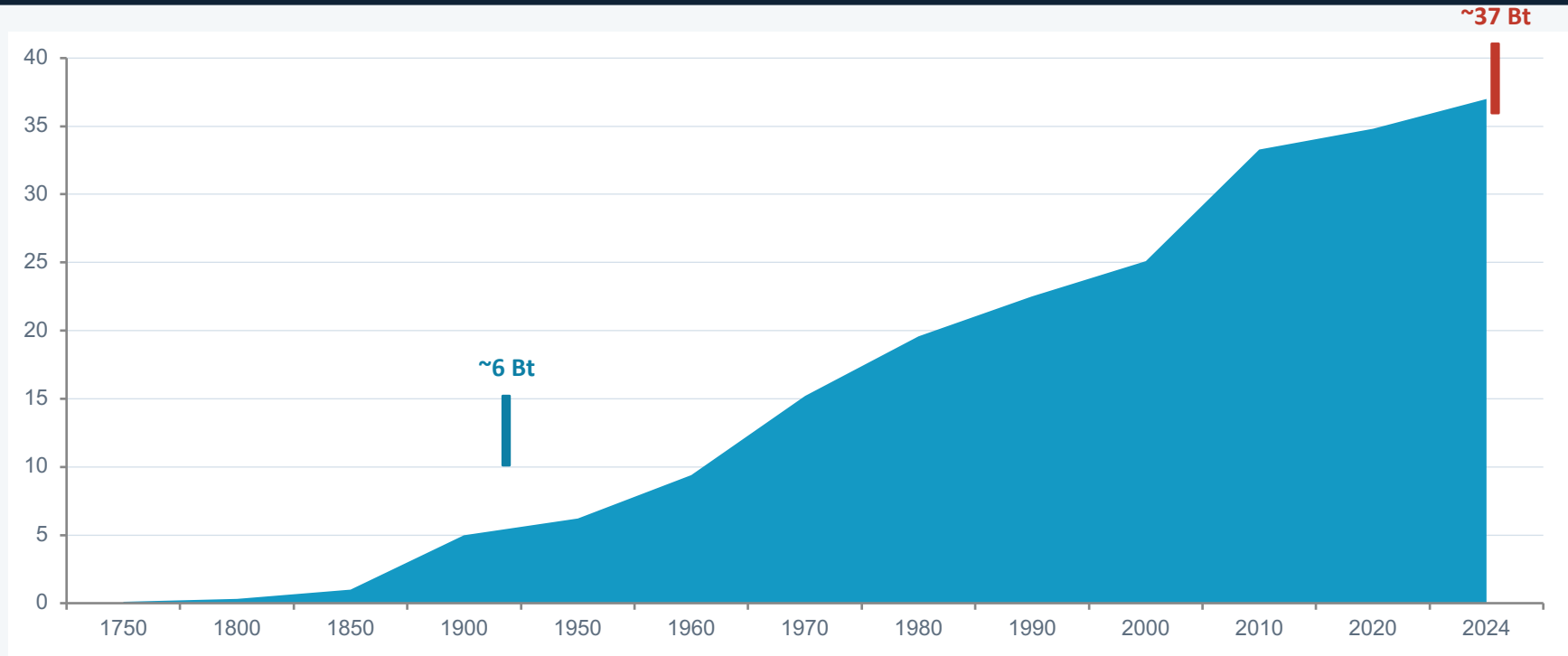
Who Emits, How Much, and What's Changing

Scope: Fossil fuels & industry · 1750–2024

Data: Global Carbon Budget 2025 via Our World in Data · Authors: Hannah Ritchie & Max Roser

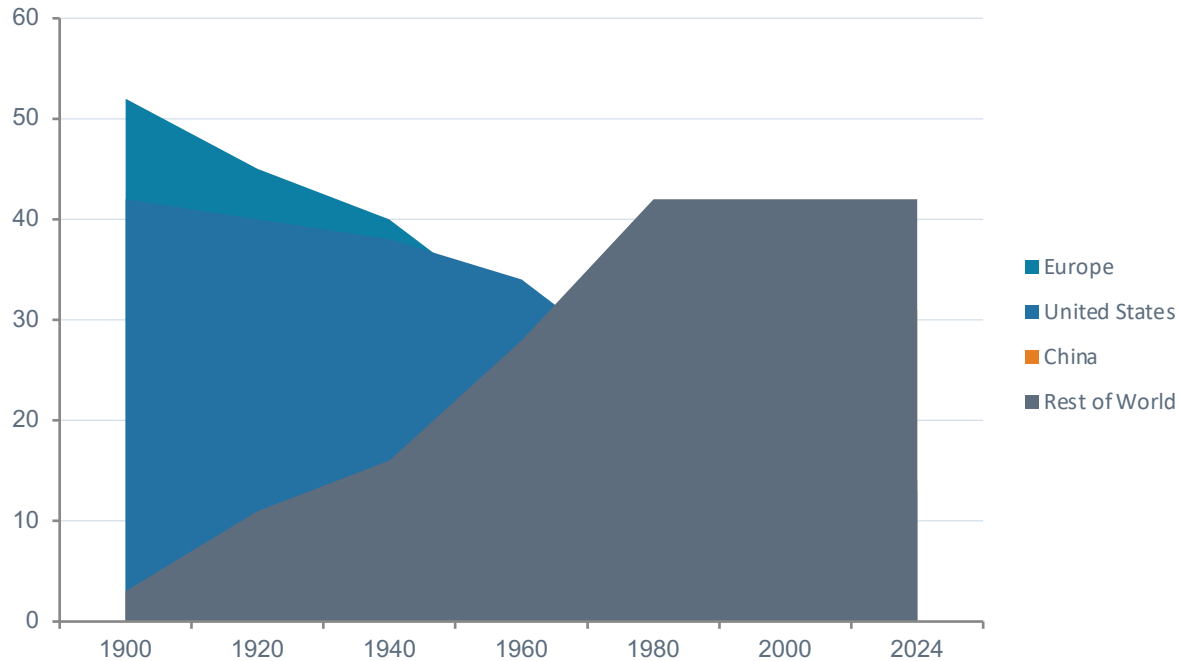
Audience: Policy analysts · Climate negotiators · Sustainability researchers

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today



Post-WWII acceleration drove a 6× increase in 70 years. Growth has slowed — but emissions have not yet peaked.

Europe and the US Fell from >90% of Emissions in 1900 to Under One-Third Today



Key Shift

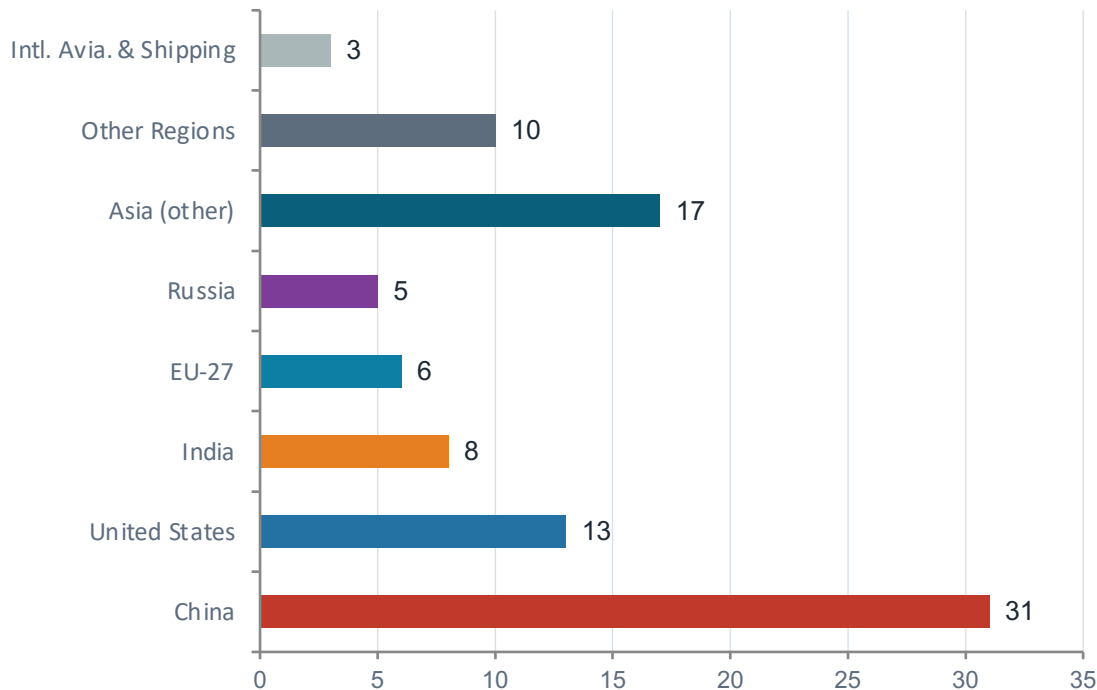
1900: Europe + US dominated at **>90%**

2024:
Europe + US = **~27%**

China alone now = **~31%**

Asia (led by China) now produces ~50% of global emissions

China Now Emits More Than One-Quarter of Global CO₂



31%

China

Largest emitter

13%

United States

2nd largest, declining

<10%

Africa + S. America

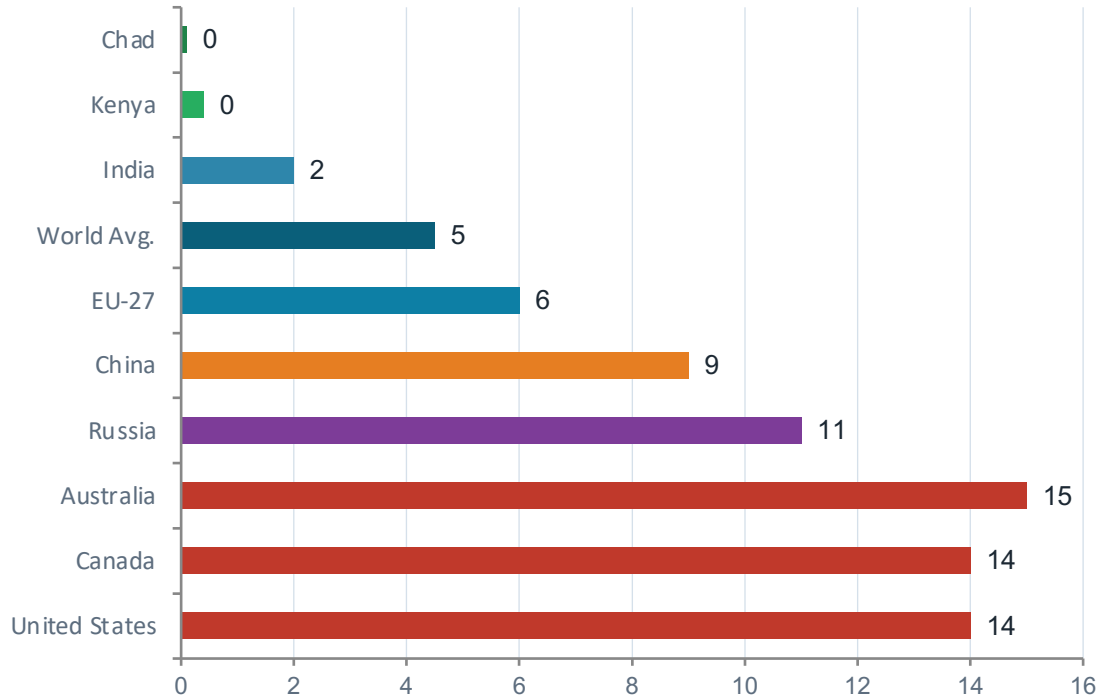
Combined share

~35 Bt

Global Total

Annual emissions

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada



150x

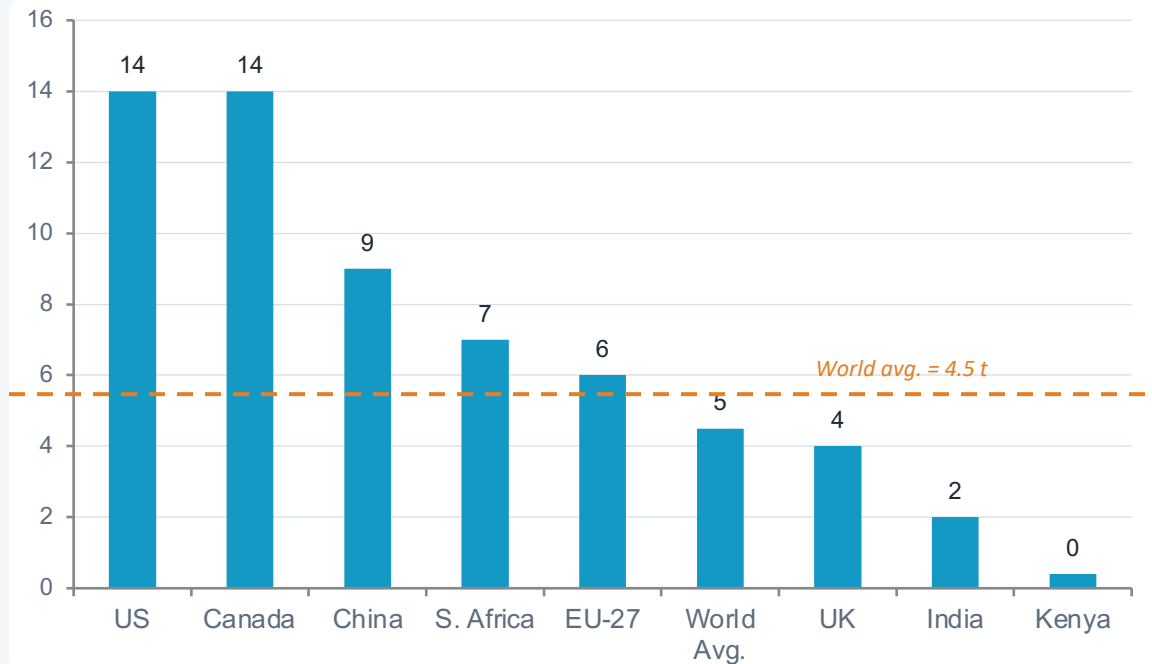
Gap between highest
and lowest emitters

*The average American emits in ~2 days what
the average person in Chad emits in a full year.*

Note

Even among wealthy nations, per capita emissions vary widely. France & the UK (~4–6 t) emit far less than the US (14 t) due to nuclear & renewable energy shares.

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India



US: 14 t

3× the world average. Highest among large economies.

China: 9 t

Largest total emitter, but per capita still 1/3 below the US.

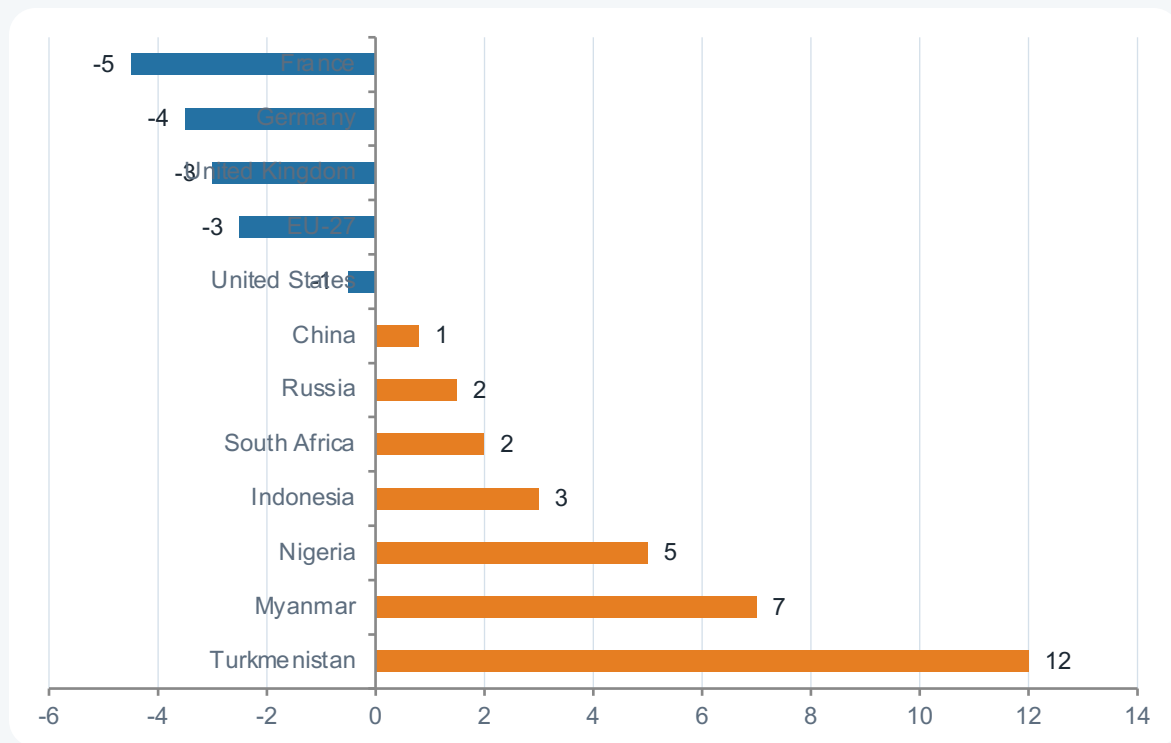
India: 2 t

Very low per capita despite being 3rd largest total emitter.

Kenya: 0.4 t

Comparable to other Sub-Saharan African nations.

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew



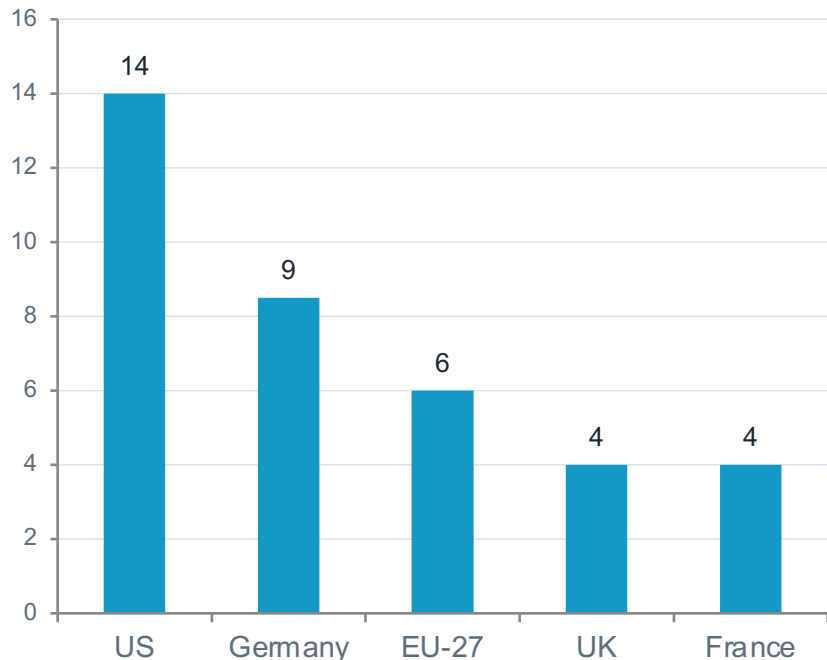
2024 Change Legend

- Emissions increased
- Emissions declined

- Declines: US, EU, UK, Germany, France — continued clean energy transition
- Increases: Russia, parts of SE Asia, Africa offset some gains
- Turkmenistan: largest increase >10% — likely gas flaring

Net result: global emissions in 2024 remained roughly flat vs. 2023 — no significant increase, but no peak yet.

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US



Why the gap?

Countries with similar prosperity levels can have very different per capita emissions depending on their energy mix.

France

~70% nuclear + renewables → ~4 t/capita

UK

~45% renewables + gas shift → ~4 t/capita

Germany

~50% fossil-based electricity → ~8.5 t/capita

US

~60% fossil-based electricity → ~14 t/capita

Energy policy and technology choices can decouple prosperity from emissions — even among wealthy nations.

Key Takeaways

Responsibility Is Shared but Unequal

1

No peak yet

Global CO₂ from fossil fuels exceed 35 Bt/yr — and have not yet peaked despite slowing growth.

2

China is #1, but per capita still below US

China alone emits ~31% of global total, yet per capita (~9 t) remains well below the US (~14 t).

3

Historical responsibility: Europe & US

Europe and the US produced >90% of cumulative emissions. Today they account for <27% of annual output.

4

150× per capita gap persists

A person in Chad (~0.1 t/yr) emits 150× less than a person in the US (~14 t/yr).

5

Energy policy can decouple prosperity & emissions

France and the UK show that wealthy nations can achieve low per capita emissions through nuclear & renewable energy.

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